

CARBON AND ITS COMPOUNDS

Teacher Lesson Guide

1. Metadata

Topic	Carbon and its Compounds
Level	Class X
Lesson Length	60 minutes

2. Learning Objectives

- LO1: Explain how carbon's tetravalency and catenation enable the formation of a large variety of compounds.
- LO2: Distinguish saturated from unsaturated carbon compounds by identifying the type of C-C bonds present.
- LO3: Identify common hydrocarbons and classify them as alkanes, alkenes, alkynes or aromatic (benzene).
- LO4: Describe the typical physical properties of hydrocarbons and relate them to molecular structure.
- LO5: Summarize the significance of Wöhler's synthesis of urea for organic chemistry.

3. Prerequisite Knowledge

- Understanding of valence electrons and covalent bonding.
- Familiarity with single, double and triple covalent bonds.
- Ability to read and write simple molecular formulas.

4. Key Terms

Term	Definition
Tetravalency	The ability of carbon to form four covalent bonds with other atoms.
Catenation	The property of carbon atoms to bond to each other, forming chains, branches, or rings.
Saturated compound	A carbon compound that contains only single C-C bonds.
Unsaturated compound	A carbon compound that contains one or more double or triple C-C bonds.
Hydrocarbon	A compound composed solely of carbon and hydrogen atoms.
Alkane	A saturated hydrocarbon containing only single C-C bonds (e.g., methane, ethane).
Alkene	An unsaturated hydrocarbon containing at least one C=C double bond (e.g., ethene).
Alkyne	An unsaturated hydrocarbon containing at least one C≡C triple bond (e.g., ethyne).
Benzene	An aromatic hydrocarbon with the formula C ₆ H ₆ , featuring a ring of six carbon atoms with alternating double bonds.

5. Key Points

- Carbon forms four covalent bonds and can link to other carbon atoms, creating countless structures.
- In 1828 Friedrich Wöhler made urea from ammonium cyanate, disproving the need for a 'vital force' in organic synthesis.
- Hydrocarbons contain only carbon and hydrogen and are grouped as alkanes, alkenes, alkynes or aromatic compounds.
- Saturated compounds have only single C-C bonds; unsaturated compounds contain double or triple bonds.
- Hydrocarbons have low melting/boiling points and are poor conductors because they are held together by weak intermolecular forces.

6. Real-World Applications

Alkanes as Everyday Fuels

Alkanes such as propane and butane are used for cooking, heating and as fuel for vehicles because they are stable, release a lot of energy on combustion, and have low boiling points that make them easy to store as liquids or gases.

Think about: Why are saturated hydrocarbons like propane suitable for use as fuels?

Expected answer: They are chemically stable, have high energy content, and their low boiling points allow them to be stored and transported as liquids or gases.

Suggested time: 3 minutes

Carbon Catenation and Plastic Polymers

The ability of carbon atoms to form long chains (catenation) underlies the creation of polymeric materials such as polyethylene, which are made from repeating alkane units.

Think about: How does carbon's capacity to link with itself lead to the formation of everyday plastics?

Expected answer: Carbon's catenation allows many carbon atoms to join into long chains; when these chains are linked together they form polymers that have the strength and flexibility needed for plastic products.

Suggested time: 4 minutes

7. Teacher Schedule

Time	Stage	Teacher Direction	Student Action
0-5 min	Introduction	Introduce the lesson topic and learning objectives. Use brief retrieval questioning to activate relevant prerequisite knowledge.	Respond to retrieval questions and review the learning objectives.
5-13 min	Teaching Phase 1	Emphasize the four valence electrons and the ability to form C-C bonds in chains, branches and rings. Use the real-world application 'Carbon Catenation and Plastic Polymers': The ability of carbon atoms to form long chains (catenation) underlies the creation of polymeric materials such as polyethylene, which are made from repeating alkane units. Ask students: How does carbon's capacity to link with itself lead to the formation of everyday plastics?	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.

Time	Stage	Teacher Direction	Student Action
13-25 min	Teaching Phase 2	Use structural formulas to show single, double and triple bonds. Use the real-world application 'Alkanes as Everyday Fuels': Alkanes such as propane and butane are used for cooking, heating and as fuel for vehicles because they are stable, release a lot of energy on combustion, and have low boiling points that make them easy to store as liquids or gases. Ask students: Why are saturated hydrocarbons like propane suitable for use as fuels?	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
25-35 min	Teaching Phase 3	Provide examples such as methane, ethane, ethene, ethyne and benzene.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
35-43 min	Teaching Phase 4	Link low melting/boiling points and non-conductivity to weak intermolecular forces.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
43-47 min	Teaching Phase 5	Highlight the disproof of the 'vital force' theory.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
47-50 min	Consolidation / Review	Use the remaining time to revisit key points, check understanding, clarify difficult concepts or practise a short retrieval question.	Review key learning and ask questions about any remaining areas of uncertainty.
50-55 min	Question Phase	Students complete Q1, Q2, Q3, Q4, Q5. Review answers and address remaining gaps in understanding.	Answer the lesson questions and review responses.
55-60 min	Reflection / Review	Review the learning objectives and identify areas requiring further review.	Review the learning objectives and identify any topic requiring further review.

8. Activities

Suggested Activities If Time Available

Molecule-Building Workshop (12 min)

Using molecular model kits, students construct methane, ethane, ethene, ethyne and benzene. They label each bond type (single, double, triple, aromatic) and discuss how carbon's tetravalency enables each structure.

Historical Debate: Vital Force vs. Laboratory Synthesis (10 min)

In small groups, students read a short excerpt on the 'vital force' theory and Wöhler's urea experiment. They then debate which view is more convincing and write a brief paragraph summarizing the significance of the 1828 synthesis.

9. Questions and Answers

Q1. What does the term 'catenation' refer to in the context of carbon chemistry?

Answer: The ability of carbon atoms to bond to each other, forming long chains, branched structures, or rings.

Q2. How can you tell if a carbon compound is saturated or unsaturated?

Answer: If it contains only single C-C bonds it is saturated; the presence of double or triple C-C bonds makes it unsaturated.

Q3. Provide the molecular formula and classification (alkane, alkene, alkyne) for methane, ethane, ethene and ethyne.

Answer: Methane: CH₄ - alkane; Ethane: C₂H₆ - alkane; Ethene: C₂H₄ - alkene; Ethyne: C₂H₂ - alkyne.

Q4. Why do most hydrocarbons have relatively low melting and boiling points compared to ionic compounds?

Answer: Because the intermolecular forces between hydrocarbon molecules are weak van der Waals forces, and there are no ions to create strong electrostatic attractions.

Q5. What was the significance of Friedrich Wöhler's synthesis of urea in 1828?

Answer: It disproved the 'vital force' theory by showing that an organic compound (urea) could be synthesized from inorganic precursors, establishing that organic chemistry is not limited to living systems.